PoE (primary side, isolated 1500 Vrms) -> isolated forward converter -> VBUS 48 V -> rails and GaN bridges

POWER AND AMPLIFIER

poe_in
PoE++ input: magjack, bridges, PD
(block A)
File: poe_in.kicad_sch

iso_dcdc
Isolated active-clamp forward, 48 V
(block B)
File: iso_dcdc.kicad_sch

rails
Secondary rails and sequencing
(block C)
File: rails.kicad_sch

amp_woofer
GaN BTL power stage: woofer
(block D)
File: amp_woofer.kicad_sch

iso_aux
Converter sync clock and VBUS droop
(block B)
File: iso_aux.kicad_sch

amp_sense
VBUS, current and temperature sensing
(block D)
File: amp_sense.kicad_sch

amp_tweeter
GaN BTL power stage: tweeter
(block D)
File: amp_tweeter.kicad_sch

COMPUTE, NETWORK, AUDIO AND I/O

Compute: LAN8720A RMII <-> ECP5 (PWM, MAC, PTP, 1-bit DAC) <-> PIO 8-bit link <-> RP2350B

ethernet
Ethernet PHY and clocks
(block E)
File: ethernet.kicad_sch

fpga
ECP5 FPGA and configuration
(block E)
File: fpga.kicad_sch

mcu
RP2350B, flash, USB
(block E)
File: mcu.kicad_sch

audio_out
1-bit DAC, headphone and line out
(block F)
File: audio_out.kicad_sch

io
User I/O, debug and test points
(block G)
File: io.kicad_sch

audio_in
Measurement mic and MEMS mic
(block F)
File: audio_in.kicad_sch

Inter-sheet signals are global labels; rails are power symbols. Net names: design/parts/*-interfaces.md

Open source, MIT. Sheets in block-diagram order: power in -> amp -> compute -> audio
PoE++ GaN speaker amplifier. Design: design/intent.md, design/architecture.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /
File: satellite.kicad_sch

Title: Studio Amp satellite, variant A

Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 1/14

PoE++ input, block A (references 100–199). VPRI = bridge output = TPS2373 VDD; VPD_N = bridge OUTN = TPS2373 VSS; VPRI_RTN = TPS2373 RTN (after the hot-swap) = block B's primary return.

CBULK (61 uF) is drawn on iso_dcdc: block B's C215 47 uF 100 V Al + C211–C213 3 x 4.7 uF at T1 (>= 10 uF for U101, <= 120 uF; inrush 12.7 ms).

VPRI

VPRI, VPRI_RTN, VB_PRI: block B (iso_dcdc)

PAIR SET A (pins 1–2 / 3–6)

PAIR SET B (pins 4–5 / 7–8)

PWR_FLAG

PWR_FLAG

TG12 4 Q101 PSMN040

TG36 4 Q103 PSMN040

TG45 4 Q105 PSMN040

TG78 4 Q107 PSMN040

POE_CT12

POE_CT36

POE_CT45

POE_CT78

BG12 4 Q102 PSMN040

BG36 4 Q104 PSMN040

BG45 4 Q106 PSMN040

BG78 4 Q108 PSMN040

VPD_N

Two ideal-diode bridges. High-side FET: drain VPRI, source CT; low-side: drain CT, source VPD_N.
PSMN040-100MSE: 25–57 mV drop at 0.865 A per pair set, 0.20 W for all eight hot (A-bridge-lt4321.md).
Layout: each pair set's 4 FETs round its two CT pins; TG/BG traces short to U101.

J101 LPJK6078AHNL

4x 1CT:1CT, Bob Smith inside

LINE SIDE (primary)

PHY SIDE (secondary)

CT1–2 13 POE_CT12

CT3–6 14 POE_CT36

CT4–5 15 POE_CT45

CT7–8 16 POE_CT78

MD3+ 3 MD3– 2 MDCT3 1

MD4+ 8 MD4– 9 MDCT4 7

MD1+ 11 MD1– 10 MD2+ 4 MD2– 5

MDCT1 12 MDCT2 6

ETH_AVDD

ETH_TXP ETH_TXN ETH_RXP ETH_RXN

LEDY+ 18

LEDG+ 20 LEDY– 17 LEDG– 19

ETH_LED_G_A

ETH_LED_Y_K

SH

GND

C101 1n 2kV X7R 1206

R101 2.0M 1% 1206

R102 2.0M 1% 1206

R103 2.0M 1% 1206

R104 2.0M 1% 1206

R105 2.0M 1% 1206

GND

SECONDARY (GND)

Isolation barrier (R–32, 1500 Vrms): this box edge, through J101

SHIELD –> GND: C101 1 nF 2 kV || R101–R105 (10 Mohm).
R–32 hipot: <= 1.2 kVpk on J101's internal 1 nF, <= 1.23 kVpk on C101, AC hipot only.
ESD +–8 kV: 1.33 kV peak, 267 V per resistor.
A-shield-network.md

MD3/MD4 windings unused (100BASE-TX uses pairs 1–2 and 3–6): their line-side CTs still carry power (CT4–5, CT7–8).

U101 LT4321IUF

POE_CT12 15 IN12 POE_CT36 2 IN36

POE_CT45 3 IN45 POE_CT78 6 IN78

OUTP 12

TG12 16 BG12 14 TG36 1 BG36 13

TG45 4 BG45 8 TG78 5 BG78 7

EN 10

EN 11

OUTN 9

VPD_N

R107 1M 1% 0603

R106 100k 1% 0603

D101 BZT52C27

Q109 BSS123

D102 BZT52C10

R108 2.2M 1% 0805

C102 1n 50V X7R

VPRI_RTN

LT4321 enable (A-bridge-lt4321.md calc 3): on for VPRI >= 32.8 V (D101, R106), held off below 21 V (R107). Q109 pulls EN low whenever the TPS2373 hot-swap is open (RTN–VSS > 2.8 V), so the bridges run only with CBULK connected (LT4321 p.7). R108 2.2 M keeps the detection signature at 24.4–24.9 k.

D103 SMCJ58A TVS

C103 100n 100V X7R 0805

R109 24.9k 1% 0603

D103: <= 93.6 V at 16 A (abs max 100 V). C103 = C_PD, the only cap on VDD–VSS (0.05–0.12 uF). Place both at U101.

U102 TPS2373-4

VDD 1 DEN 2 CLSA 3 CLSB 6 REF 7 MPS_DUTY 9 NC 20 VSS 4

VC_OUT 14 VC_IN 15 PG 13 TPH 18 TPL 17 BT 19 AMPS_CTL 8 UVLO_SEL 16 APD 10 RTN 11

PD_PG

R110 63.4R 1% 0805

R111 90.9R 1% 0805

R112 49.9k 1% 0603

R113 1.3k DNP

VB_PRI

C104 1u 50V X7R 0805

C105 100n 50V X7R

VC_OUT unused (1 uF only). VC_IN = VB_PRI, 14.3 V from block B (TPS2373 7.3.7).

VISO_P5V

R114 10k 1% 0603

R115 10k 1% 0603

R116 10k 1% 0603

Class B: CLSA 63.4R, CLSB 90.9R (TPS2373 Table 1). RDEN 24.9k (trim to 24.3k if the signature reads high). AMPS_CTL: R113 DNP, auto-MPS off (R–30). MPS_DUTY, UVLO_SEL open: APD to RTN.
A-pd-tps2373.md, A-pd-passives.md

U104 78L05

VB_PRI

C107 330n 50V X7R 0805

IN 3 GND 2 OUT 1

VISO_P5V

C108 100n 50V X7R

U104 78L05: SOT-89 O-G-I pinout is the –AB3-R order only.
VISO_P5V (primary): <= 15.3 mA, 177 mW at VB_PRI 14.8 V.

VISO_P5V

R117 10k 1% 0603

UCC_SS 1 Q110 2N7002

VPRI_RTN

PD_PG_N LOW = SS/SD above the 2N7002 threshold (1–2.5 V): PD out of inrush and UCC2897A running. UCC_SS is block B's SS/SD node after its 1 k from PD_PG; the gate draws <= 80 nA.

SECONDARY (GND, +3V3)

VISO_P5V

+3V3

C109 100n 50V X7R

C110 100n 50V X7R

U103 ISO7740DWR

VCC1 16 INA 3 INB 4 INC 5 IND 6 NC 7 GND1 8

VCC2 1 OUTA 15 OUTB 14 OUTC 13 OUTD 11 GND2EN2 10 GND 9

PD_BT_N PD_TPL_N PD_TPH_N PD_PG_N

All four active LOW. ISO7740 (no F suffix) reads HIGH while the primary is unpowered. TPH/TPL/BT valid 20–28 ms after PG releases.
Truth table: A-pd-tps2373.md

Primary = VPRI / VPD_N / VPRI_RTN; secondary = GND. Barrier R–32: 1500 Vrms (dashed).
Block A. Audits: design/parts/A–*.md (values, calculations, datasheet pages)

Sequoia Alexander, github.com/sequoia–hope/studio–amp

Sheet: /poe_in/
File: poe_in.kicad_sch

Title: Studio Amp satellite: PoE++ input: magjack, bridges, PD

Size: A2	Date: 2026–11–08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 2/14

PRIMARY: VPRI_RTN

ISOLATION BARRIER 1500 Vrms (R-32)

SECONDARY: GND

T1 wired 1:2: primary N1 || N2 = 4 T (Lm 25 uH).
secondary N3 + N4 in series (10-8) = 8 T.
Dots = winding starts (B-transformer.md).
Layout: N1 and N2 symmetric.

N5 aux, 4 T: forward pulse - VPRI, averaged
-> LINEOV: output OVP at VBUS = 53.3 V,
independent of the opto (B-controller.md)

PRIMARY INPUT CAPACITORS

VPRI from the PD (block A), 38.4-57 V, <= 1.85 A.
C215 47u: LTA321 needs >= 10 uF; damps the cable.
Layout: C211-C214 at T1 pins 2/3 and Q201's Rcs,
the 737 kHz pulse-current loop (1.34 A rms).
B-primary-power.md

OUTPUT FILTER, IBUS SHUNT 30 mR, VBUS BULK

L202 68u: 0.35-0.55 A p-p, 1.65 A peak at CC.
C231-C236 6 x 4.7u 100V ahead of the shunt keep 737 kHz ripple out of it. B-output.md

VBUS 47.9 V +-1.5 %: 6 x 1000u 63V KZN (C247 DNP footprint).
OVP: SMCJ51A + LINEOV 53.3 V + gateware 55 V (agreed with block D)

T201
750343576 (1:2)

UCC2897A CONTROLLER, PRIMARY STAGE

VIN: 100k from VPRI (JFET start-up unused;
VDD from VB_PRI). R219 DNP option.

LINEUV: on 40.4 V,
off 38.4 V

LINEOV: R209 470k from node A;
23.7k || 47n averages (1 ms)

RDEL 4.53k = 50 ns. RON 31.6k / ROFF 14.3k:
free-run 656 kHz (590-708), synced to
737.28 kHz. DMAX 0.68-0.715. RSLOPE 316k.
CSS 1u: 60 ms soft start. PD_PG holds SS
low during PD inrush: UCC_SS -> block A.
GND (8) and PGND (13) meet only at the IC
(datasheet 11.1). B-controller.md

Active clamp, P-FET low side:
Vc = 100 V on C209 || C210 (250 V X7R).
B-primary-power.md

FORWARD SR

FREEWHEEL SR

Both SRs: RC snubber 22p + 47R 2512; RCD-Z clamp (US1G, 10n, SMBJ130A) holds the
drain and NCP4306 CS <= ~165 V (200 V abs max). MIN_TON 750R - 125 ns, MIN_TOFF 2k - 256 ns.
Layout: NCP4306 GND Kelvin to its SR source pad. CS to the drain pad. B-sync-rect.md

C223
2.2n Y2/X1 250VAC 2220

C224
2.2n Y2/X1 250VAC 2220 DNP

C223 Y2 2.2n at T1: interwinding
displacement current. C224 DNP
(EMI tuning). B-feedback.md

IBUS SENSE (NA281)

INA281A4 200 V/V x 30 mR = 6.0 V/A:
100R + 470n kills 737 kHz ripple.
IBUS_SENSE 2.0 V/A, 67k source -> RP2350
ADC4 (block E fits 100 nF). B-feedback.md

OPTO LED, CV LOOP (TL431) AND CC LOOP (TL431, ILIM_SEL)

FB: 4.99k to VREF, 6.8n (pole 4.7 kHz).
Opto gain CTR x 4.99k / 499R = 10.
fc = 1 kHz, PM = 66 deg (B-feedback.md)

CV: 2.495 V x (1 + 47.5k/2.61k) = 47.9 V.
Type II: 115k + 4.7n (zero 294 Hz).

CC limit (ILIM_SEL1:0, reset 00 via 100k):
00 0.44 A, 01 0.94 A, 10 0.80 A, 11 1.30 A.
Integrator 100k + 10n. B-feedback.md

VB_PRI BIAS BUCK (PRIMARY)

VB_PRI 14.3 V (13.9-14.8), <= 85 mA: UCC2897A, block A's TPS2373
VC_IN and isolator, the sync clock (iso_aux). Starts at VPRI 30.3 V.
600 kHz (RON 267k): Type III ripple Rr 200k, Cr 1n, Cac 100n.
B-bias.md

Active-clamp forward, 737.28 kHz synced, T1 1:2, VPRI 38.4-57 V -> VBUS 47.9 V
Block B. Audits: design/parts/B-*.md (overview B-converter-design.md)
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /iso_dcdc/
File: iso_dcdc.kicad_sch

Title: Studio Amp satellite: Isolated active-clamp forward, 48 V

Size: A2	Date: 2026-11-08	Rev: A
KiCad E.D.A. 9.0.8		Id: 3/14

PRIMARY SYNC CLOCK (VPRI_RTN) -> UCC2897A SYNC

VB_PRI
U252
HT7550-1 5V
VIN VOUT GND
C258 1u 25V X7R 0603
C259 1u 25V X7R 0603
VPRI_RTN
V5_CLK

V5_CLK 5 V, <= 10 mA: oscillator, counter, one-shot (0.1 W in the SOT-89)

C262 100n 50V X7R 0603
U253
74LVC1GU04
GND VCC
R258 1k 1% 0603
C260 33p 50V C0G 0603
C261 33p 50V C0G 0603
Y251
7.3728MHz 20pF
R257 1M 1% 0603

7.3728 MHz, CL 20 pF, +-50 ppm all-in (SXMH7.3728F2020F30, -40..85 C):
33p + 33p + ~5 pF stray = 21.5 pF.
Rf 1M, Rs 1k (drive ~70 uW).

U254
74HC4017
CLK CKEN Reset VDD VSS Cout Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9
C263 100n 50V X7R 0603

Only this gate runs from VREF, so SYNC <= VREF + 0.3 V in every state; 10k holds SYNC low when VREF is off (IOFF output). VREF budget <= 0.6 mA of 5 mA. B-sync-clock.md

U255A
74LVC1G123
B A CLR Cext RCext Q
C264 470p 50V C0G 0603
R259 10k 1% 0603

U255B
74LVC1G123
VCC GND
C265 100n 50V X7R 0603

C266 100n 50V X7R 0603
VREF_ACF
U256
74LVC1G08
GND VCC
R260 100 1% 0603
R261 10k 1% 0603
ACF_SYNC
CLK_OK

Q1: one 136 ns pulse per 10 clocks = 737.28 kHz (< (1 - DMAX) x T = 0.39 us). CP1 and MR low.

Clock-OK: retriggered by CO; Rext 10k / Cext 470p -> tw ~ 4.5 us (> 1.36 us CO period, <= 10 us). Clock stops -> CLK_OK low within ~4.5 us -> SYNC low -> UCC2897A free-runs (590-708 kHz). B triggers on A falling edges with B held high.

Nothing crosses here: the clock stays primary, the detector and buffer are secondary.

ISOLATION BARRIER

VSR 10 V BIAS BUCK (SECONDARY)

PWR_FLAG
VBUS
C252 100n 100V X7R 0603
C251 1u 100V X7R 0805
R251 280k 1% 0603
U251
LM5019
VIN RON ULVO BST SW FB VCC RTN RTNPAD
C256 1n 50V C0G 0603
R256 82k 1% 0603
C254 10n 50V X7R 0603
C257 100n 50V X7R 0603
L251 220u 0.36A
C253 1u 25V X7R 0603
R254 51.1k 1% 0603
R255 10k 1% 0603
R252 71.5k 1% 0603
R253 10k 1% 0603
C255 10u 25V X7R 1206
VSR
GND

VSR 10 V (9.7-10.3), <= 37 mA: NCP4306s, TL431s, opto LED, INA2B1, droop comparator. Starts at VBUS 7.48 V (51.1k/10k), dropout below ~10.6 V. 400 kHz (RON 280k); Type III Rr 82k, Cr 1n, Cac 100n. B-bias.md

ACF_PULSE BUFFER (+3V3)

+3V3
DRV_FWD
R270 10k 1% 0603
R271 4.99k 1% 0603
U259B
74LVC2G08
ACF_PULSE
U259A
74LVC2G08
C269 100n 50V X7R 0603
U259C
74LVC2G08
GND

DRV_FWD (10 V forward-SR gate) / 3 -> 3.33 V -> gate B (+3V3, 5.5 V-tolerant inputs) -> ACF_PULSE: one edge per conduction pulse, 737.28 kHz locked. FPGA B6 via block E's 4.7k; clock-fault rule in B-interfaces.md. Gate A unused.

VBUS DROOP DETECTOR

VSR
R262 3.9k 1% 0603
VBUS
R263 150k 0.1% 0805
R264 10.0k 0.1% 0603
U258
TL431B
VREF_D
R265 10k 1% 0603
R266 866k 1% 0603
C267 33n 50V X7R 0603
VBD
U257A
LM393
R267 3.3k 1% 0603
R268 10k 1% 0603
R269 15k 1% 0603
Q251
2N7002
POE_LOSS_N
U257C
LM393
C268 100n 50V X7R 0603
U257B
LM393

POE_LOSS_N low when VBUS < 39.5 V (38.5-40.5), released > 41.1 V (40.0-42.2): 150k/10k 0.1 %, 33n: tau 0.31 ms. 2N7002 via 10k/15k (gate 0.42 V max when off). Pull-up 4.7k to +3V3 at the FPGA (block E). B-vbus-droop.md

737.28 kHz sync clock, VSR bias, VBUS-droop detector, ACF_PULSE
Block B. Audits: design/parts/B-sync-clock, B-bias, B-vbus-droop, B-interfaces
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /iso_aux/
File: iso_aux.kicad_sch

Title: Studio Amp satellite: Converter sync clock and VBUS droop

Size: A3 | Date: 2026-11-08 | **Rev: A**
KiCad E.D.A. 9.0.8 | Id: 4/14

5 V BUCK LM5013 ~213 kHz (C-5v-buck.md)

VBUS R301 NR 1206 — PWR_FLAG — TP301 TP VBUS

C301 4u7 100V — C302 4u7 100V — C303 100n 100V — GND

R302 1M 0805 — R303 52.3k — R304 64.9k — GND

U301 LM5013

VIN — BST — C304 2n2 — SW — L301 47u SRP1265A — SW_5V — D301 SS2A10 — GND

EN/UVLO — FB — R306 100k — R305 34.8k 0.5% — FB_5V — GND

RON — PGOOD — PWR_FLAG — PG_5V

GND — EP — LM5013

UVLO on 30.2 V / off 28.2 V

C305 22u 25V — C306 22u 25V — C307 22u 25V — GND

PWR_FLAG — +5V_BUCK — TP302 +5V_BUCK

R307 330k 0805 — C308 3n3 — C309 330p C0G — FB_5V

OR R308 1206

Hot loop C301–U301–D301 tight.
FB parts at pin 5, away from SW.
SW_5V: pad + GND spring site (lo)

5 V SOURCE SELECT TPS2116
buck has priority (C-5v-mux.md)

VUSB — PWR_FLAG — C311 4u7 25V — GND

R309 56k — C310 1u 50V — PG_5V — R310 100k — GND

U302 TPS2116

VIN1 — VIN2 — VOUT — TP303 TP +5V — +5V

MODE — PR1 — ST — C312 4u7 — C313 470u 10V — GND

R311 100k — +3V3 — PWR_FROM_POL

3V3 BUCK TPS62A02A 2.4 MHz FPWM

3V3 = 0.6 x (1 + 100k/22.0k) = 3.33 V
0.1 %: ECP5 VCCIO max 3.465 V

+5V — PWR_FLAG — JP301 bridged — C314 4u7 25V — GND

U303 TPS62A02A

VIN — SW — L302 1u SRP4020TA — PWR_FLAG — +3V3

EN — FB — R313 100k 0.1% — R312 220k 0.1% — GND

PG — GND — +3V3 — TP304 TP +3V3 — GND

+3V3 — R314 100k — PG_3V3

1V1 BUCK TPS62A02A (ECP5 VCC)

1V1 = 0.6 x (1 + 84.5k/100k) = 1.107 V
400 mA design; starts ~1 ms after 3V3

+5V — PWR_FLAG — JP302 bridged — C316 4u7 25V — GND

U304 TPS62A02A

VIN — SW — L303 1u SRP4020TA — PWR_FLAG — +1V1

EN — FB — R316 84.5k — R315 100k — GND

PG — GND — +1V1 — C317 22u 10V — C318 22u 10V — GND

TP305 TP +1V1

2V5 LDO TPS7A9401 (ECP5 VCCAUX, C-2v5-ldo.md)

VOUT = 150 uA x 16.7k = 2.505 V
!EN_UV = PG_3V3: with 1V1, after 3V3
!SNS: Kelvin to the bead input

+3V3 — PWR_FLAG — JP303 bridged — C319 1u 25V — GND

U305 TPS7A9401

IN — OUT — SNS — R318 715k 1% — C320 10u 25V — +2V5 — TP306 TP +2V5

EN_UV — FB_PG — PG — R319 68.1k — GND

PG_3V3 — R317 16.7k 0.1% — C321 1n 100V — GND

NR/SS — GND — EP — GND

GATE-DRIVE SUPPLY VDRV 7.0 V LM5017 + TLV6703 (C-vdrv-buck.md)

VBUS R320 NR 1206 — PWR_FLAG — TP307 TP VBUS

C322 1u 100V — C323 2u2 100V — C324 100n 100V — GND

VDRV_EN — R323 10k — R321 191k 0805 — R322 100k — GND

U306 LM5017

VIN — BST — C326 20n — SW — L304 47u SRN6045TA — SW_VDRV — PWR_FLAG

RON — FB — R326 330k 0805 — C329 10u 50V — C330 10u 25V — GND

ULVO — VCC — C325 1u 50V — GND

RTN — RTNPAD — GND — VDRV

R325 46.4k — C327 100n — C328 1n — R324 10k — GND — VDRV

VDRV — TP308 TP VDRV — R327 150k — C331 100n — GND — R329 4.7k

U307 TLV6703

SENSE — GND — OUT — R328 10k — R330 2.7k — VDRV_PG

VDRV_EN low = off (100k pull-down).
Firmware enables after FPGA DONE.
PD_PG_N low and POE_LOSS_N high.
VDRV 7.01–7.12 V typ (type III offset)

POWER-GOOD BUFFERS
ECP5 inputs never load PG nets

+3V3 — R331 4.7k — PG_5V_N

PG_5V — Q301 2N7002 — GND

+3V3 — R332 4.7k — PG_3V3_N

PG_3V3 — Q302 2N7002 — GND

Active low: low = rail good.
>= 2.6 V vs the 150 uA ECP5
pull-down when not good

3V3_Q LDO TPS7A9401 (DAC reclock, C-3v3q-ldo.md)

+5V — JP304 bridged — PWR_FLAG — C332 4u7 25V — GND

+5V — R333 47k — R334 10k — GND

U308 TPS7A9401

IN — OUT — TP309 TP +3V3_Q — +3V3_Q

EN_UV — FB_PG — R336 931k — C334 22u 10V — C335 22u 10V — GND

NR/SS — GND — PG — R337 68.1k — GND

EP — GND — R335 20k 0.1% — C333 4u7 25V — GND

+3V3 — R338 100k — PG_3V3Q

SNS: Kelvin trace to the 74LVC74 row (block F)
COUT ceramic, ESR <= 20 mOhm

3V3_A LDO TPS7A2033P
(block D sense reference, C-3v3a-ldo.md)

+5V — PWR_FLAG — JP305 bridged — C336 1u 10V — GND

U309 TPS7A2033P

IN — OUT — +3V3_A — TP310 TP +3V3_A

EN — GND — NC — C337 4u7 25V — GND

Always on with +5V: block D's ADCs are
powered whenever the FPGA drives them

+-15 V PRE-REGULATOR LM5017 FLY-BUCK (C-pm15-flybuck.md)

VBUS R339 NR 1206 — PWR_FLAG — TP311 TP VBUS

C338 1u 100V — C339 2u2 100V — C340 100n 100V — GND

R341 124k 0805 — R340 350k 0805 — R342 5.36k — GND

U310 LM5017

VIN — BST — C342 20n — SW — L305 — PWR_FLAG — +18V_PRE — TP312 pad +18V_PRE

RON — FB — SW_AN — R345 390k 0805 — C341 1u 50V — GND

ULVO — VCC — GND

RTN — RTNPAD — GND

68u 1:1 PSSCA-0704-68M

C345 10u 50V — C346 10u 50V — GND

UVLO on 29.8 V / off 27.3 V (124k / 5.36k)

+18V_PRE — R344 137k — C343 100n — R343 10k — GND

C344 1n — +18V_PRE — D302 B1100 — PWR_FLAG — −18V_PRE — TP313 pad −18V_PRE

R346 10R 2W — C347 220p C0G 100V — C348 10u 50V — C349 10u 50V — R347 10k — GND

Windings 1–3 primary, 2–4 secondary (dots at 1, 2).
Secondary dot end -> D cathode, non-dot -> GND:
rectified output is negative. 10k preload.

+15 V / −15 V LDOs TPS7A4901 / TPS7A3001 (C-pm15-ldos.md)

FB301 BLM18PG221 — PWR_FLAG — C350 10u 50V — GND

U311 TPS7A4901

IN — NC — DNC — C351 10n — GND

EN — OUT — FB — R348 15k — C352 10n — R349 9.76k — GND

NR/SS — GND — EP — GND — +15V — C353 10u 50V — TP314 TP +15V — GND

+15V = 1.188 x (1 + 113k/9.76k) = 14.94 V
120 mA design incl. +5V_A, 200 mA fault

FB302 BLM18PG221 — PWR_FLAG — C354 10u 50V — GND

U312 TPS7A3001

IN — NC — DNC — C355 10n — GND

EN — OUT — FB — R350 15k — C356 10n — R351 10k — GND

NR/SS — GND — EP — GND — −15V — C357 10u 50V — TP315 TP −15V — GND

−15V = −1.179 x (1 + 118k/10k) = −15.09 V
DNC (7) unconnected; PowerPAD to GND

+5V_A LDO TPS7A4901
(PCM1808 analog, mic bias)

+15V — C358 10u 25V — GND

U313 TPS7A4901

IN — NC — DNC — C359 10n — GND

EN — OUT — FB — R352 32.4k — C360 10n — C361 4u7 25V — +5V_A — TP316 TP +5V_A — GND

NR/SS — GND — EP — GND — R353 10k — GND

+5V_A = 1.188 x (1 + 32.4k/10k)
= 5.04 V, <= 15 mA, 15 uVrms

SEQUENCING (C-sequencing.md)

VBUS >= 30.2 V: LM5013 -> +5V_BUCK -> PG_5V
-> TPS2116 selects the buck (else VUSB)
+5V -> 3V3 -> PG_3V3 -> 1V1, 2V5 (~1 ms)
VBUS >= 29.8 V: Fly-Buck -> +-15 V (14 ms)
-> +5V_A: +15V >= 7.1 V enables 3V3_Q
VDRV off until VDRV_EN (firmware, after
FPGA DONE, PD_PG_N low, POE_LOSS_N high)
USB only: +5V, 3V3, 1V1, 2V5, 3V3_A

Hold-up >= 34 ms after the VBUS < 40 V
trip at 60 V on 6 mF (UVLO-off <= 30.2 V)

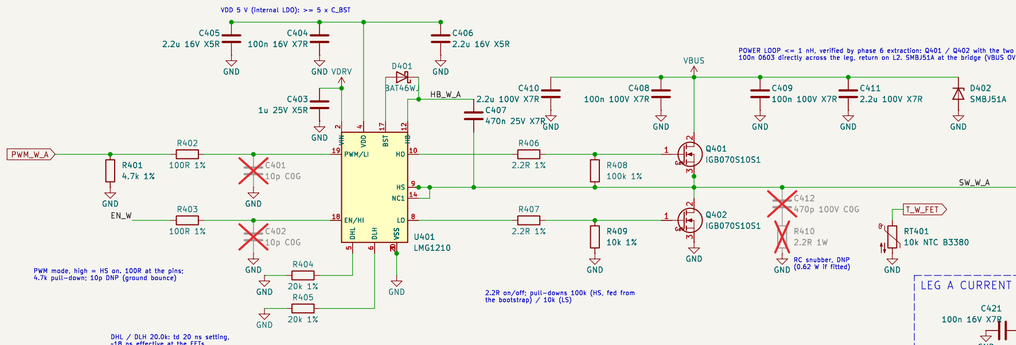
R-80: OR / bridged-jumper link at every
converter input; test point on every rail;
SW-node pads with GND springs on lo sheet

Block C. Audits: design/parts/C-*.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp
Sheet: /rails/
File: rails.kicad_sch

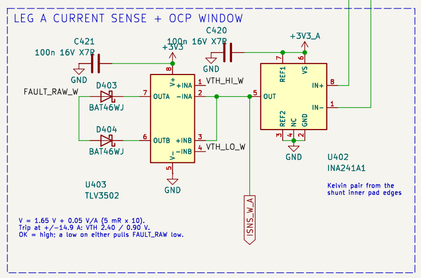
Title: Studio Amp satellite: Secondary rails and sequencing

Size: A2	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 5/14

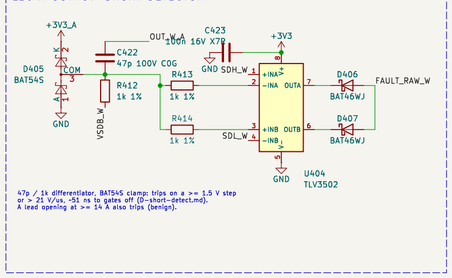
LEG A



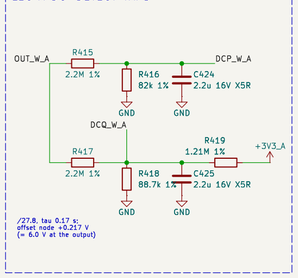
LEG A CURRENT SENSE + OCP WINDOW



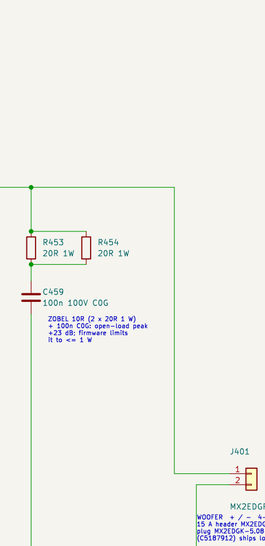
LEG A OUTPUT-SHORT DETECTOR



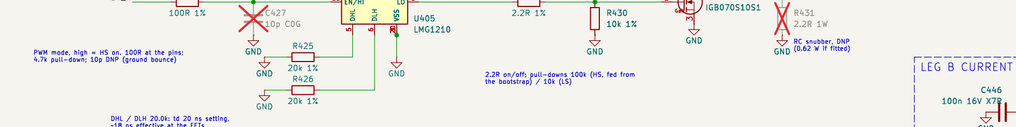
LEG A DC-DETECT TAPS



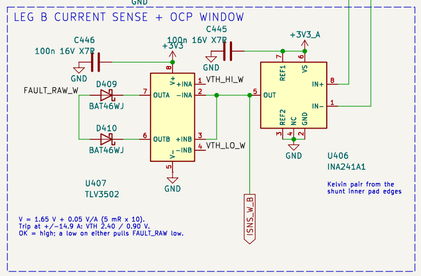




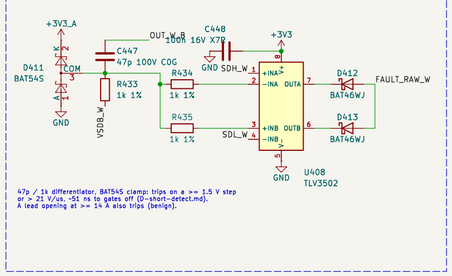
LEG B



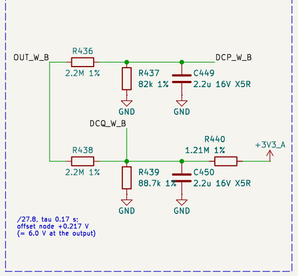
LEG B CURRENT SENSE + OCP WINDOW



LEG B OUTPUT-SHORT DETECTOR

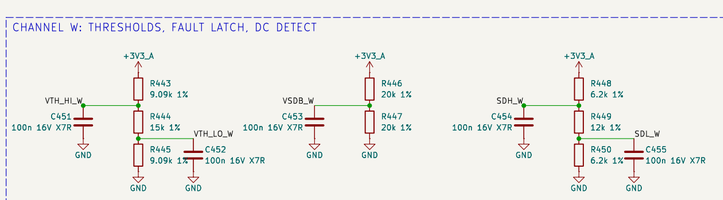


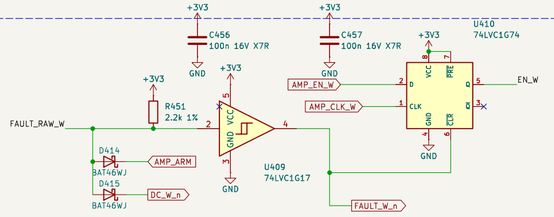
LEG B DC-DETECT TAPS

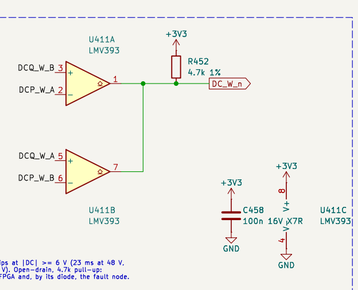




CHANNEL W: THRESHOLDS, FAULT LATCH, DC DETECT







gen/amp.py, both channels; the tweeter adds the 2 x 22 uF PP HPF caps
Block D. Audits: design/parts/D-*.md (IGB070S10S1, fsw 576 kHz, D-interfaces.md)
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /amp_woofer/
File: amp_woofer.kicad_sch

Title: Studio Amp satellite: GaN BTL power stage: woofer

Size: A2	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 6/14

LEG A

VDD 5 V (internal LDO): >= 5 x C_BST

C505 2.2u 16V X5R
C504 100n 16V X7R
C506 2.2u 16V X5R
GND
GND VDRV
C503 1u 25V X5R
D501 BAT46WJ
HB_T_A
C510 100n 100V X7R
C508 100n 100V X7R
VBUS
C509 100n 100V X7R
C511 2.2u 100V X7R
D502 SMBJ51A
C507 470n 25V X7R
2.2u 100V X7R

POWER LOOP <= 1 nH, verified by phase 6 extraction: Q501 / Q502 with the two 100n 0603 directly across the leg, return on L2. SMBJ51A at the bridge (VBUS 0V scheme).

PWM_T_A
R501 4.7k 1%
R502 100R 1%
C501 10p C0G
U501 LMG1210
PWM/LI
EN/HI
DHL
DLH
VDD
BST
HO
HS
NC1
LO
VSS
R503 100R 1%
EN_T
C502 10p C0G
R504 20k 1%
R505 20k 1%
R506 2.2R 1%
R507 2.2R 1%
R508 100k 1%
R509 10k 1%
Q501 IGB070S10S1
Q502 IGB070S10S1
C512 470p 100V C0G
R510 2.2R 1W
RC snubber, DNP (0.62 W if fitted)
SW_T_A
T_T_FET
RT501 10k NTC B3380
L501 10u SER2915H
R511 5m 1% 1W

PWM mode, high = HS on, 100R at the pins; 4.7k pull-down; 10p DNP (ground bounce)

DHL / DLH 20.0k: td 20 ns setting, ~18 ns effective at the FETs

2.2R on/off; pull-downs 100k (HS, fed from the bootstrap) / 10k (LS)

LC filter: 10 uH + 7 x 100n C0G to GND per side: f0 60 kHz, Q 0.53 / 1.06 at 4 / 8 ohm

C513 100n 100V C0G
C514 100n 100V C0G
C515 100n 100V C0G
C516 100n 100V C0G
C517 100n 100V C0G
C518 100n 100V C0G
C519 100n 100V C0G
OUT_T_A

LEG A CURRENT SENSE + OCP WINDOW

C521 100n 16V X7R
+3V3
C520 100n 16V X7R
+3V3_A
D503 BAT46WJ
D504 BAT46WJ
FAULT_RAW_T
U503 TLV3502
V+ +INA -INA OUTA OUTB +INB -INB V-
VTH_HI_T
VTH_LO_T
U502 INA241A1
REF1 REF2 VS IN+ IN- OUT NC GND
ISNS_T_A
Kelvin pair from the shunt inner pad edges

V = 1.65 V + 0.05 V/A (5 mR x 10).
Trip at +/-14.9 A: VTH 2.40 / 0.90 V.
OK = high; a low on either pulls FAULT_RAW low.

LEG A OUTPUT-SHORT DETECTOR

+3V3_A
D505 BAT54S
A K COM
C522 47p 100V C0G
R512 1k 1%
VSDB_T
OUT_T_A
C523 100n 16V X7R
GND SDH_T
R513 1k 1%
R514 1k 1%
SDL_T
+3V3
U504 TLV3502
+INA -INA OUTA +INB -INB OUTB
D506 BAT46WJ
D507 BAT46WJ
FAULT_RAW_T

47p / 1k differentiator, BAT54S clamp: trips on a >= 1.5 V step or > 21 V/us, ~51 ns to gates off (D-short-detect.md). A lead opening at >= 14 A also trips (benign).

LEG A DC-DETECT TAPS

OUT_T_A
R515 2.2M 1%
R516 82k 1%
C524 2.2u 16V X5R
DCP_T_A
DCQ_T_A
R517 2.2M 1%
R518 88.7k 1%
R519 1.21M 1%
C525 2.2u 16V X5R
+3V3_A

/27.8, tau 0.17 s; offset node +0.217 V (= 6.0 V at the output)

OUT_T_A
R520 140k 0.1%
PF_T_A
R521 10k 0.1%
Post-filter sense /15, this film 0.1 %, to the AD58353 (amp_sense)

PROTECTIVE HPF: 2 x 22u MKP in parallel = 44 uF; 904 / 603 / 452 Hz at 4 / 6 / 8 ohm; blocks DC if a leg fails short (R-5s)

C561 22u 450V MKP
C560 22u 450V MKP
OUT_T_HPF
R553 20R 1W
R554 20R 1W
C559 100n 100V C0G
ZOBEL 10R (2 x 20R 1 W) + 100n C0G: open-load peak +23 dB; firmware limits it to <= 1 W

J501
MX2EDGRC-5.08-02P
TWEETER + / - , 4-8 ohm
15 A header MX2EDGRC-5.08; plug MX2EDGR-5.08-02P (C5187912) ships loose

LEG B

VDD 5 V (internal LDO): >= 5 x C_BST

C530 2.2u 16V X5R
C529 100n 16V X7R
C531 2.2u 16V X5R
GND
GND VDRV
C528 1u 25V X5R
D508 BAT46WJ
HB_T_B
C535 100n 100V X7R
C533 100n 100V X7R
VBUS
C534 100n 100V X7R
C536 2.2u 100V X7R
C532 470n 25V X7R
2.2u 100V X7R

POWER LOOP <= 1 nH, verified by phase 6 extraction: Q503 / Q504 with the two 100n 0603 directly across the leg, return on L2.

PWM_T_B
R522 4.7k 1%
R523 100R 1%
C526 10p C0G
U505 LMG1210
PWM/LI
EN/HI
DHL
DLH
VDD
BST
HO
HS
NC1
LO
VSS
R524 100R 1%
EN_T
C527 10p C0G
R525 20k 1%
R526 20k 1%
R527 2.2R 1%
R528 2.2R 1%
R529 100k 1%
R530 10k 1%
Q503 IGB070S10S1
Q504 IGB070S10S1
C537 470p 100V C0G
R531 2.2R 1W
RC snubber, DNP (0.62 W if fitted)
SW_T_B
T_T_IND
RT502 10k NTC B3380
L502 10u SER2915H
R532 5m 1% 1W

PWM mode, high = HS on, 100R at the pins; 4.7k pull-down; 10p DNP (ground bounce)

DHL / DLH 20.0k: td 20 ns setting, ~18 ns effective at the FETs

2.2R on/off; pull-downs 100k (HS, fed from the bootstrap) / 10k (LS)

LC filter: 10 uH + 7 x 100n C0G to GND per side: f0 60 kHz, Q 0.53 / 1.06 at 4 / 8 ohm

C538 100n 100V C0G
C539 100n 100V C0G
C540 100n 100V C0G
C541 100n 100V C0G
C542 100n 100V C0G
C543 100n 100V C0G
C544 100n 100V C0G
OUT_T_B

LEG B CURRENT SENSE + OCP WINDOW

C546 100n 16V X7R
+3V3
C545 100n 16V X7R
+3V3_A
D509 BAT46WJ
D510 BAT46WJ
FAULT_RAW_T
U507 TLV3502
V+ +INA -INA OUTA OUTB +INB -INB V-
VTH_HI_T
VTH_LO_T
U506 INA241A1
REF1 REF2 VS IN+ IN- OUT NC GND
ISNS_T_B
Kelvin pair from the shunt inner pad edges

V = 1.65 V + 0.05 V/A (5 mR x 10).
Trip at +/-14.9 A: VTH 2.40 / 0.90 V.
OK = high; a low on either pulls FAULT_RAW low.

LEG B OUTPUT-SHORT DETECTOR

+3V3_A
D511 BAT54S
A K COM
C547 47p 100V C0G
R533 1k 1%
VSDB_T
OUT_T_B
C548 100n 16V X7R
GND SDH_T
R534 1k 1%
R535 1k 1%
SDL_T
+3V3
U508 TLV3502
+INA -INA OUTA +INB -INB OUTB
D512 BAT46WJ
D513 BAT46WJ
FAULT_RAW_T

47p / 1k differentiator, BAT54S clamp: trips on a >= 1.5 V step or > 21 V/us, ~51 ns to gates off (D-short-detect.md). A lead opening at >= 14 A also trips (benign).

LEG B DC-DETECT TAPS

OUT_T_B
R536 2.2M 1%
R537 82k 1%
C549 2.2u 16V X5R
DCP_T_B
DCQ_T_B
R538 2.2M 1%
R539 88.7k 1%
R540 1.21M 1%
C550 2.2u 16V X5R
+3V3_A

/27.8, tau 0.17 s; offset node +0.217 V (= 6.0 V at the output)

OUT_T_B
R541 140k 0.1%
PF_T_B
R542 10k 0.1%
Post-filter sense /15, this film 0.1 %, to the AD58353 (amp_sense)

CHANNEL T: THRESHOLDS, FAULT LATCH, DC DETECT

+3V3_A
R543 9.09k 1%
R544 15k 1%
R545 9.09k 1%
VTH_HI_T
VTH_LO_T
C551 100n 16V X7R
C552 100n 16V X7R
GND

OCP thresholds 2.396 / 0.904 V (9.09k / 15k / 9.09k, ratiometric)

R546 20k 1%
R547 20k 1%
VSDB_T
C553 100n 16V X7R

Short-detector bias 1.65 V (2 x 20k)

R548 6.2k 1%
R549 12k 1%
R550 6.2k 1%
SDH_T
SDL_T
C554 100n 16V X7R
C555 100n 16V X7R

Short-detector window 2.65 / 0.65 V (6.2k / 12k / 6.2k)

+3V3
C556 100n 16V X7R
C557 100n 16V X7R
U509 74LVC1G17
VCC GND
R551 2.2k 1%
FAULT_RAW_T
D514 BAT46WJ
D515 BAT46WJ
AMP_ARM
DC_T_n
U510 74LVC1G74
D CLK PRE CLR Q
AMP_EN_T
AMP_CLK_T
EN_T
FAULT_T_n

Fault latch: any OK low (OCP x 4, output short x 4, DC, AMP_ARM low) pulls FAULT_RAW low; Schmitt -> CLR -> Q (EN) low: both LMG1210 off in <= 216 ns (ramp) / ~51 ns (short). The FPGA reads FAULT_n and re-arms with AMP_EN = 1 + an AMP_CLK edge, PRE tied high.

U511A LMV393
DCQ_T_B
DCP_T_A
R552 4.7k 1%
DC_T_n
U511B LMV393
DCQ_T_A
DCP_T_B
U511C LMV393
C558 100n 16V X7R

DC detect: trips at |DC| >= 6 V (23 ms at 48 V, 50 ms at 24 V). Open-drain, 4.7k pull-up; DC_n to the FPGA and, by its diode, the fault node.

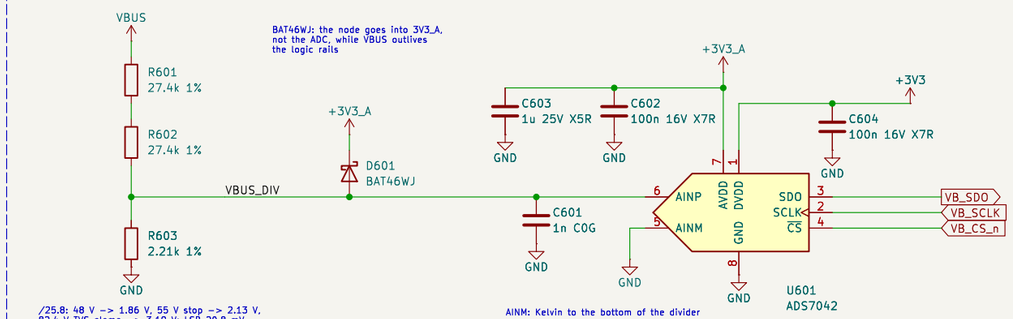
gen/amp.py, both channels; the tweeter adds the 2 x 22 uF PP HPF caps
Block D. Audits: design/parts/D-*.md (IGB070S10S1, fsw 576 kHz, D-interfaces.md)
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /amp_tweeter/
File: amp_tweeter.kicad_sch

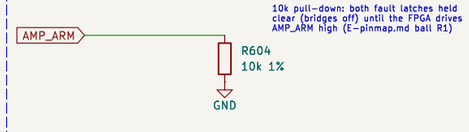
Title: Studio Amp satellite: GaN BTL power stage: tweeter

Size: A2	Date: 2026-11-08	Rev: A
KiCad E.D.A. 9.0.8		Id: 7/14

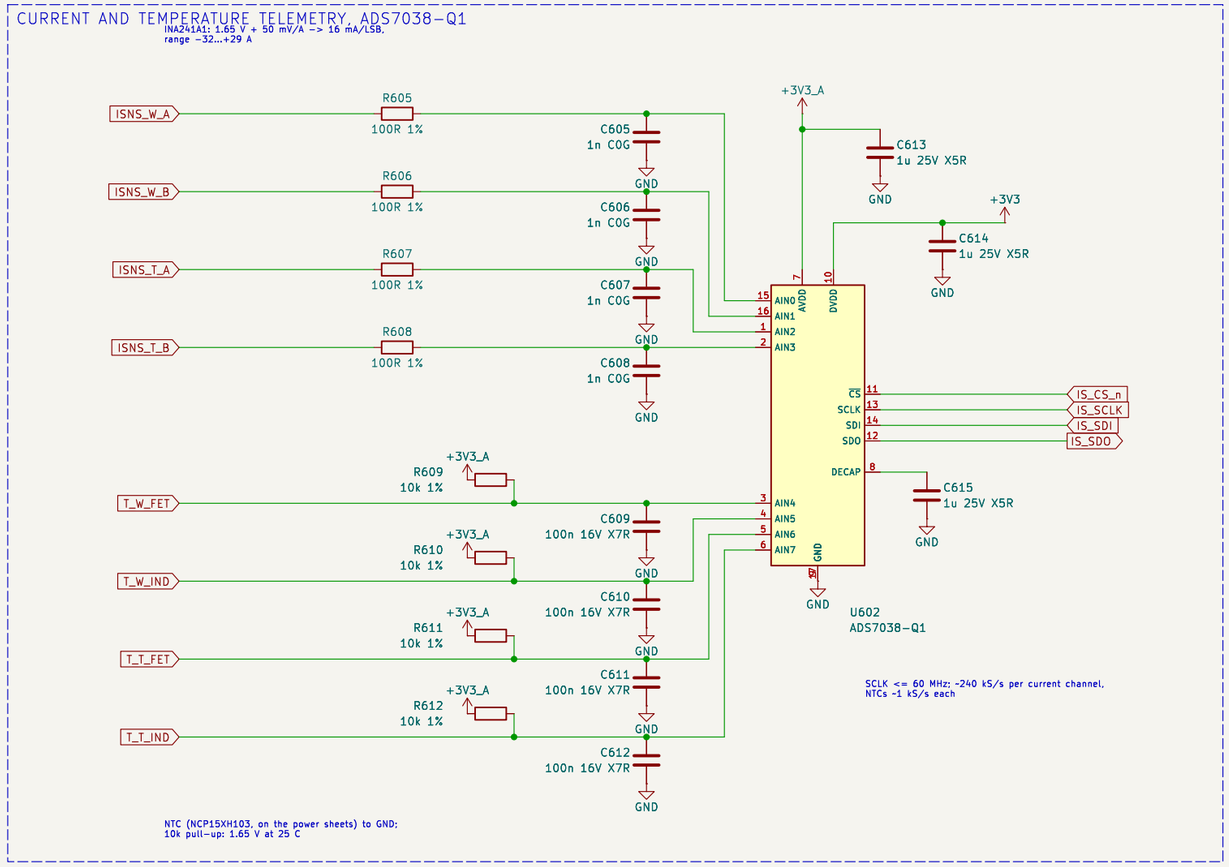
VBUS SENSE, ADS7042 (FEED-FORWARD, POE-LOSS RULE)



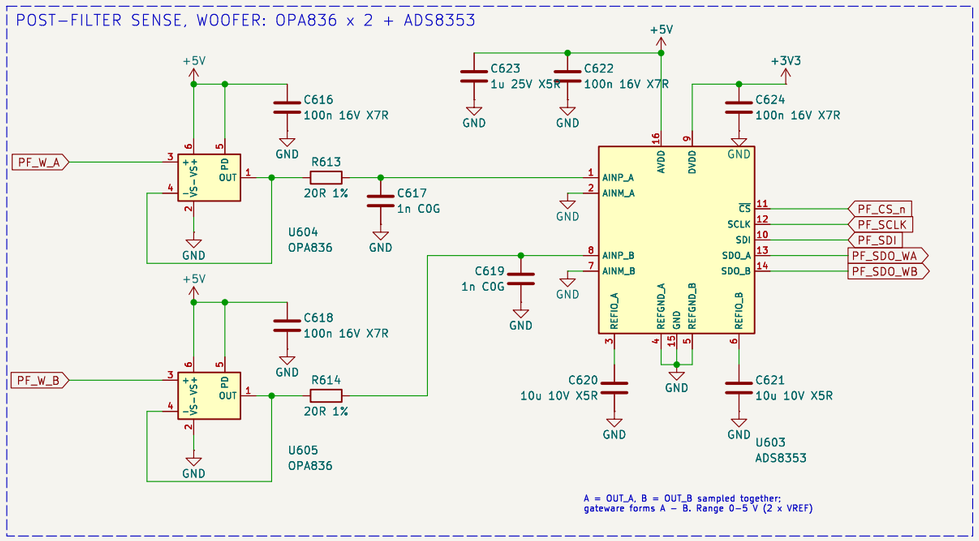
AMP_ARM DEFAULT



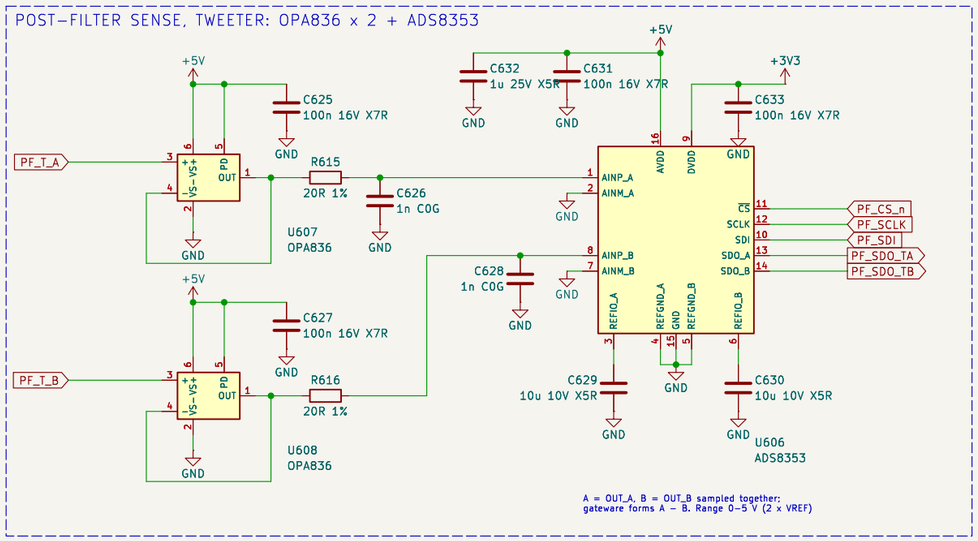
CURRENT AND TEMPERATURE TELEMETRY, ADS7038-Q1



POST-FILTER SENSE, WOOFER: OPA836 x 2 + ADS8353



POST-FILTER SENSE, TWEETER: OPA836 x 2 + ADS8353



PF dividers (140k/10k 0.1%) sit on the power sheets; shared CS/SCLK/SDI, one SDO per ADC channel (D-postfilter-sense.md). +5V load ~24 mA.

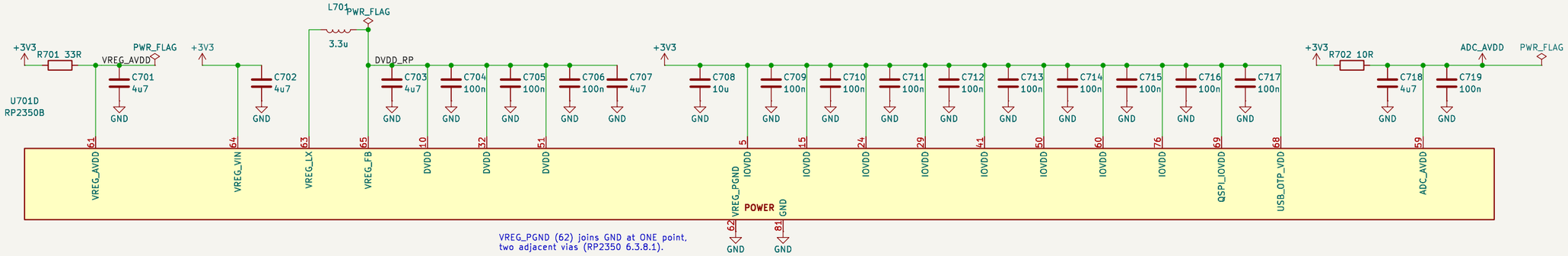
+3V3_A (block C LDO): ADC and INA241 reference
Block D. Audits: D-vbus-adc.md, D-isense-adc.md, D-ntc.md, D-postfilter-sense.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /amp_sense/
File: amp_sense.kicad_sch

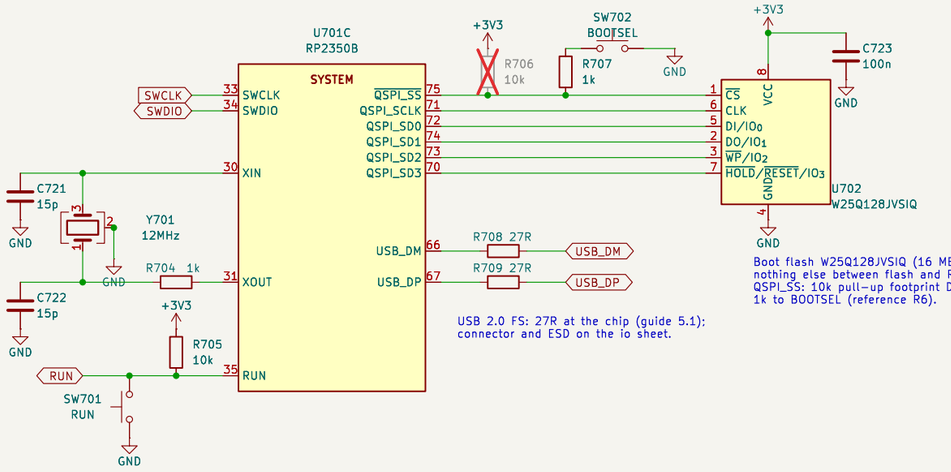
Title: Studio Amp satellite: VBUS, current and temperature sensing

Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 8/14

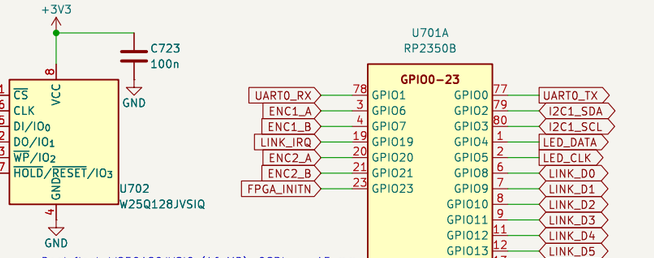
Core regulator, RP2350 6.3.8 / Minimal board: CIN, L and COUT on the chip's side; copy the reference corner (chips/rp2350.md). L: AOTA-B201610S3R3 3.3u, dot to DVDD_RP, reference rotation (orientation changes Vout). DVDD_RP is the RP2350's own 1.1 V (internal regulator); it is not the FPGA's +1V1.



VREG_PGND (62) joins GND at ONE point, two adjacent vias (RP2350 6.3.8.1).



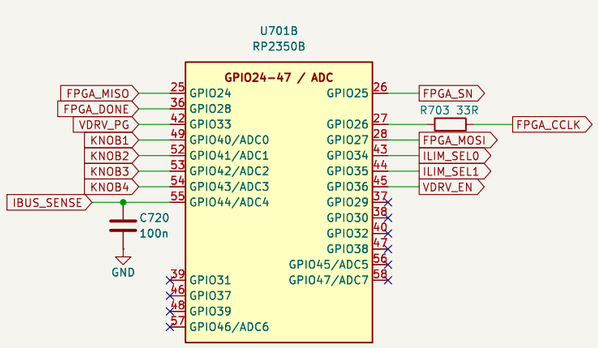
ABM8-272-T3 12 MHz with 15p + 15p and 1k at XOUT: Raspberry Pi's tuned network (guide 4); traces < 5 mm.



Boot flash W25Q128JVSIQ (16 MB). QSPI <= 15 mm, nothing else between flash and RP2350. QSPI_SS: 10k pull-up footprint DNP (reference R1); 1k to BOOTSEL (reference R6).

USB 2.0 FS: 27R at the chip (guide 5.1); connector and ESD on the io sheet.

PIO link to the ECP5 (E-link.md): 8-bit, 37.5 MHz, 22R series at each driver is in the FPGA design rule; LINK_* tri-stated while the FPGA is configured. ENC_*: encoder header on the io sheet (block G pulls).



SSPI configuration (SPI1): FPGA_CCLK 33R at the RP2350, pad set to 8 mA (1k pull-up at the FPGA). Reads <= 18.75 MHz, bitstream writes 37.5 MHz. IBUS_SENSE: 100n at the pin (B's 67k source, 24 Hz). ILIM_SEL0/1, VDRV_EN: reset Hi-Z, pulled down at blocks B and C (safe state).

Regulator, decoupling, crystal and flash copied from the Raspberry Pi RP2350B Minimal board
Block E. Audits: design/parts/E-mcu-rp2350b.md, E-mcu-passives.md, E-xtal-12mhz.md, E-flash-

Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /mcu/
File: mcu.kicad_sch

Title: Studio Amp satellite: RP2350B, flash, USB

Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 9/14

CORE: VCC +1V1 (1.045–1.155 V). 3 x 10u + 100n per ball; VCCAUX +2V5 through a 120 R bead, 10u + 100n per ball (TN-02038 Tab 3.1).
I/O: every VCCIO on +3V3 (max 3.465 V). 10u per bank (22u on bank 6, the PWM bank) + 100n per ball. Layout: 100n under the ball field between the dog-bone vias; bulk just outside the package.

+1V1
C801 10u
C802 10u
C803 10u
C804 100n
C805 100n
C806 100n
C807 100n
C808 100n
C809 100n
+2V5
PWR_FLAG
VCCAUX
FB801 BLM18PG121
C810 10u
C811 100n
C812 100n
CORE POWER
U801H
LFE5U-25F-7BG256I

+3V3
C813 10u
C814 100n
C815 100n
C816 10u
C817 100n
C818 100n
C819 10u
C820 100n
C821 100n
C822 10u
C823 100n
C824 100n
C825 22u
C826 100n
C827 100n
C828 10u
C829 100n
C830 100n
C831 10u
C832 100n
I/O POWER (all banks 3.3 V)
U801I
LFE5U-25F-7BG256I

U801A
LFE5U-25F-7BG256I
BANK 0 (top)
ACF_PULSE
PD_BT_N
PD_PG_N
PD_TPH_N
PD_TPL_N
PG_3V5Q
PG_3V3_N
PG_5V_N
POE_LOSS_N
R801 4k7
R802 1k
R803 1k
R804 1k
R805 1k
R806 1k
R807 1k
R808 1k
R809 1k
R810 4k7
+3V3
PT24A/GR_PCLK0_1
PT24B/GR_PCLK0_0
PT27A/PCLKT0_1
PT27B/PCLKC0_1
PT29A/PCLKT0_0
PT29B/PCLKC0_0

Status from always-powered parts (A, B, C):
1k in series (hot socket, DS-02012 3.6).
POE_LOSS_N: block B open drain, 4k7 pull-up
on the net (pre-config sweep, E-pinmap.md).
ACF_PULSE: 4k7 series (B's 74LVC2G08).

U801B
LFE5U-25F-7BG256I
BANK 1 (top)
PHY_NINT
RMII_CRSDV
RMII_REFCLK
RMII_RXD0
RMII_RXD1
PHY_MDC
PHY_MDIO
PHY_NRST
RMII_TXD0
RMII_TXD1
RMII_TXEN
PT35A/PCLKT1_0
PT33A/PCLKT1_1
PT33B/PCLKC1_1
PT38A/GR_PCLK1_0
PT35B/PCLKC1_0
PT38B/GR_PCLK1_1

RMII to the LAN8720A (ethernet sheet): REFCLK on
PCLKT1_0 (CB). MODE straps (RXD0/1, CRS_DV),
MDIO, nRST and nINT pulls sit at the PHY.
LPF: PULLMODE=UP on A11, A12, B11.

U801C
LFE5U-25F-7BG256I
BANK 2 (right)
LED_FAULT_N
QWIIC_FAULT_N
VUSB_DET
R811 1k
R812 1k
R813 1k
DISP_AUX7
DISP_AUX8
DISP_BLK
DISP_BL_PWM
DISP_CS
DISP_DC
DISP_RES
DISP_SCL
DISP_SDA
LED_EN
PTP_PPS
QWIIC_PWR_EN
PR23C/PCLKT2_0
PR2C
PR2B/SO_IN
PR2D
PR5D
PR8C
PR8D
PR11A
PR11B
PR14C/VREF1_2
PR17A
PR17C
PR20C/GR_PCLK2_0
PR23B/PCLKC2_1
PR20A/GR_PCLK2_1
PR23A/PCLKT2_1

Display, LED and Qwiic power (block G/C).
Always-driven inputs: 1k series,
pair partners K15, D14, F14 left open.
B15 (SO_IN) unused.

U801D
LFE5U-25F-7BG256I
BANK 3 (right)
HP_DET
HP_FAULT_N
LINK_CLK
LINK_DIR
LINK_STB
MADC_DOUT
PDM_DAT
R814 1k
R815 1k
R816 330R
R817 330R
DAC_L0
DAC_L1
DAC_L2
DAC_L3
DAC_R0
DAC_R1
DAC_R2
DAC_R3
HP_EN
LO_EN
MADC_BCK
MADC_LRCK
PDM_CLK
PR29A/GR_PCLK3_0
PR26C/PCLKT3_0
PR38C
PR38D
PR26A/PCLKT3_1
PR38A
PR26B/PCLKC3_1
PR26D/PCLKC3_0
PR29B
PR29D
PR35C
PR44C
PR47A
PR47D/LRC_GPLLOC_IN
PR35B/VREF1_3
PR29C/GR_PCLK3_1

Audio (block F) and PIO link control (RP2350).
MADC_DOUT, PDM_DAT: 330R; HP_FAULT_N, HP_DET: 1k;
partners L15, P14, P12, K12 left open.
LINK_CLK on PCLKT3_0 (M16).

U801F
LFE5U-25F-7BG256I
BANK 7 (left)
CLK_AUX
TP801
DC_T_n
DC_W_n
R818 1k
R819 1k
PL23C/PCLKT7_0
PL20A/GR_PCLK7_1
PL23A/PCLKT7_1
PL20C/GR_PCLK7_0
PL23B/PCLKC7_1
PL23D/PCLKC7_0

CLK_AUX (K1, PCLKT7_0): spare clock input
on a test pad (external low-jitter clock option).
DC_W_n / DC_T_n: 1k series, partners J2 / H2 open.

U801E
LFE5U-25F-7BG256I
BANK 6 (left)
FAULT_T_n
FAULT_W_n
IS_SDO
PF_SDO_TA
PF_SDO_TB
PF_SDO_WA
PF_SDO_WB
VB_SDO
VCXO_CLK
R820 330R
R821 330R
AMP_ARM
AMP_CLK_T
AMP_CLK_W
AMP_EN_T
AMP_EN_W
IS_CS_n
IS_SCLK
IS_SDI
PF_CS_n
PF_SCLK
PF_SDI
PWM_T_A
PWM_T_B
PWM_W_A
PWM_W_B
VB_CS_n
VB_SCLK
VCXO_TUNE
PL26C/PCLKT6_0
PL26A/PCLKT6_1
PL38B
PL44D
PL47B
PL29D
PL44C
PL35C
PL47C/LLC_GPLLOT_IN
PL26B/PCLKC6_1
PL26D/PCLKC6_0
PL41D
PL47A
PL47D/LLC_GPLLOC_IN
PL35B/VREF1_6
PL29A/GR_PCLK6_0
PL29C/GR_PCLK6_1

Power stage (block D). PWM_* from ODDRX2F at 589.824 Mb/s
(PLL x12 from VCXO_CLK on P6, LLC_GPLLOT_IN: P5 open).
FAULT_*: always driven, 330R, partners L2 / M2 open.
T3 open: it pairs with PWM_T_B (R4).
ADC chip selects: 4k7 pull-ups keep the ADCs
deselected through configuration.

U801G
LFE5U-25F-7BG256I
BANK 8 sysCONFIG + JTAG
FPGA_CCLK
FPGA_MOSI
FPGA_SN
FPGA_PROGRAMN
FPGA_TCK
FPGA_TDI
FPGA_TMS
R825 1k
R826 4k7
R827 4k7
R828 4k7
R829 4k7
R830 4k7
R831 4k7
R832 4k7
FPGA_MISO
FPGA_INITN
FPGA_DONE
FPGA_TDO
LINK_D0
LINK_D1
LINK_D2
LINK_D3
LINK_D4
LINK_D5
LINK_D6
LINK_D7
LINK_IRQ
R833 4k7
R834 4k7
R835 4k7
CCLK/MCLK/SCK
PB11B/D0/MOSI
PB13A/SN/CSN
PB15A/HOLDN/DI
PROGRAMN
CFG_0
CFG_1
CFG_2
TCK
TDI
TMS
PB11A/D1/MISO
INITN
DONE
TDO
PB9B/D2/IO2
PB9A/D3/IO3
PB6B/D4/MOSI2
PB6A/D5/MISO2
PB4B/D6/IO6
PB4A/D7/IO7
PB13B/CS1N
PB15B/DOUT/CSON
PB18A/WRITEN

sysCONFIG slave SPI from the RP2350 (SPI1, 37.5 MHz):
CFGMDN[2:0] = 001: CFG0 4k7 up, CFG1/CFG2 to GND.
CCLK 1k pull-up to VCCIO8 (TN-02038 Tab 6.2);
33R series at the RP2350 (mcu sheet).
PROGRAMN, INITN, DONE, SN, HOLDN: 4k7 up.
JTAG: TDI, TMS, TDO 4k7 up, TCK 4k7 down
(TN-02038 Tab 6.1); header on the io sheet.
LINK_D0–7 / LINK_IRQ: PIO link (E-link.md).

ECP5 LFE5U-25F-7BG256I, slave SPI from the RP2350 (CFGMDN 001). All VCCIO 3.3 V.
Block E. Audits: design/parts/E-fpga-ecp5.md, E-fpga-ecp5-passives.md; nets: E-pinmap.md

Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /fpga/
File: fpga.kicad_sch

Title: Studio Amp satellite: ECP5 FPGA and configuration

Size: A2	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 10/14

AUDIO / PWM MASTER CLOCK, 24.576 MHz VCXO

VCXO_TUNE
R917 100k
R918 100k
R919 0R
C913 1u
C914 1u
R920 0R
VCXO_VDD
PWR_FLAG
FB902 BLM18PG221
+3V3
C915 100n
C916 4u7
Vc VCXOUT VDD GND
Y902 24.576MHz VCXO
VCXO_CLK
R921 33R
MCLK_Q

Tune: 1-bit delta-sigma from ECP5 N4, 2 x (100k, 1u): 1.59 Hz poles,
> 16 bits below 2 Hz; firmware clamps Vc to 0.3-3.0 V.
Fixed-XO fallback on the same land: fit the 0R to VDD, remove the series 0R.

VCXO_CLK -> ECP5 P6 (LLC_GPLL0T_IN), shortest branch, no R.
MCLK_Q -> block F's SN74LVC2G34 (MCLK_Q).
VCXO within 10 mm of P6, RC filter beside Vc,
nothing switching under the 1u caps.

Straps (sampled at nRST release):
MODE[2:0] = 111 (all capable, auto-neg):
4k7 pull-ups beat the ECP5's 150 uA
pre-config pull-down (review MAJOR-2):
gateware: PULLMODE=UP, nRST released
>= 25 ms after the rails.
PHYAD0 = 0 (RXER internal pull-down).

nRST 10k down: PHY held in reset
until the FPGA design releases it.
MDIO 1k5 up (tau 30 ns).

+3V3
C906 4u7
C905 100n
PWR_FLAG
VDDCR
C904 470p
C903 1u

CT bypass, place at
J101 pins 12 and 6

ETH_AVDD
C910 100n
C911 100n

ETH_AVDD: analog 3.3 V island (VDD1A, VDD2A, 4 x 49R9,
magnetics centre taps, yellow LED anode) behind a
BLM18PG221 (<= 130 mA, 13 mV). VDDCR: internal 1.2 V
regulator output, 1u || 470p at pin 6 (Tab 2-7).

C907 100n
C908 100n
C909 4u7
FB901 BLM18PG221

R901 4k7
R902 4k7
R903 4k7
R904 10k
R905 1k5
R906 10k
R907 10k

RMII_RXD0
RMII_RXD1
RMII_CRSDV
RMII_TXD0
RMII_TXD1
RMII_TXEN
TP901 RXER
PHY_MDC
PHY_MDIO
PHY_NINT
PHY_NRST

U901 LAN8720AI

Pin	Name
6	VDDCR
9	VDDIO
19	VDD1A
1	VDD2A
8	RXD0/MODE0
7	RXD1/MODE1
11	CRS_DV/MODE2
10	RXER/PHYAD0
17	TXD0
18	TXD1
16	TXEN
13	MDC
12	MDIO
14	INT/REFCLKO
15	RST
5	XTAL1/CLKIN
4	XTAL2
25	VSS

R908 49R9 1%
R909 49R9 1%
R910 49R9 1%
R911 49R9 1%
TXP 21 ETH_TXP
TXN 20 ETH_TXN
RXP 23 ETH_RXP
RXN 22 ETH_RXN
C901 100n
C902 100n

49R9 + 100n at the MDI pins, symmetric per pair;
pairs 100 R differential, no vias, over solid GND.
J101 (poe_in): TX -> 11/10, RX -> 4/5.

R912 270R
R913 270R
R914 12k1 1%
LED1/REGOFF 3 ETH_LED_G_A
LED2/INTSEL 2 ETH_LED_Y_K
RBIAS 24

LED1 (link, active high) -> green 20/19:
LED2 (100M, active low) <- yellow 18/17 from VDD2A.
Straps: REGOFF = 0 (internal 1.2 V on).
nINTSEL = 1 (REF_CLK in). RBIAS 12k1 1%.

Y901 50MHz
C912 100n
R915 33R
R916 33R
RMII_REFCLK

RMII REFERENCE CLOCK: 50 MHz +-50 ppm CMOS XO, OE tied high.
Two point-to-point branches, 33R each at the XO:
PHY XTAL1 and ECP5 C8 (PCLKT1_0), <= 25 mm,
matched +-5 mm, over solid GND.

LAN8720AI in REF_CLK-in mode (nINTSEL = 1), MODE 111, PHYAD 0
Block E. Audits: design/parts/E-phy-lan8720.md, E-phy-passives.md, E-clk-rmii-50mhz.md, E-clk-

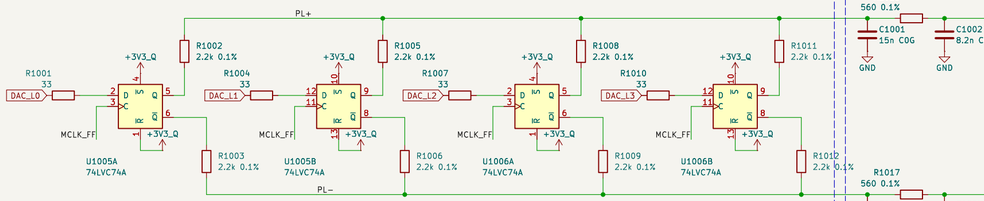
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /ethernet/
File: ethernet.kicad_sch

Title: Studio Amp satellite: Ethernet PHY and clocks

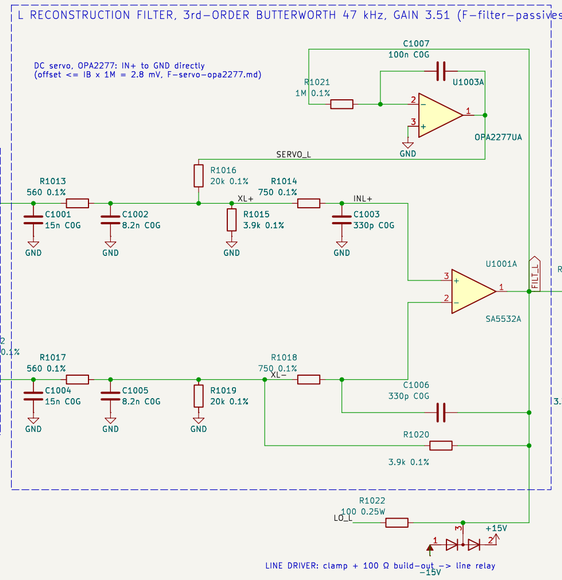
Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 11/14

L 1-BIT DAC: 4-TAP FIR, 74LVC74 ON +3V3_Q, CLOCKED BY MCLK_FF (F-dac-74lvc74.md)

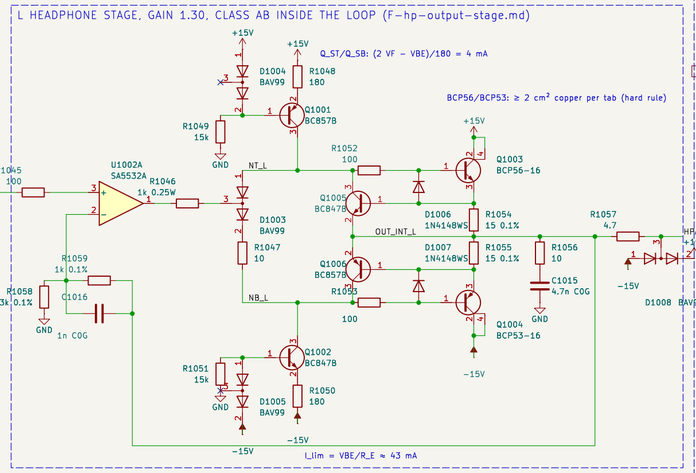


R_T 2.2k x 4 per side = 550 Ω; FIR zeros at 1.536 and 3.072 MHz

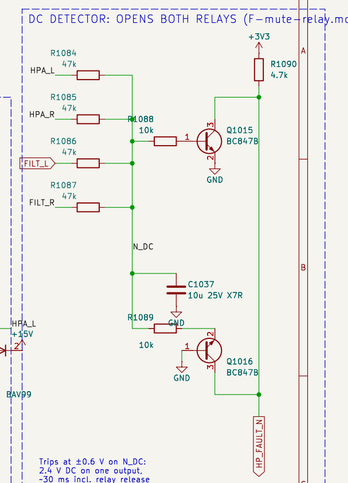
L RECONSTRUCTION FILTER, 3rd-ORDER BUTTERWORTH 47 kHz, GAIN 3.51 (F-filter-passives.md)



L HEADPHONE STAGE, GAIN 1.30, CLASS AB INSIDE THE LOOP (F-hp-output-stage.md)



DC DETECTOR: OPENS BOTH RELAYS (F-mute-relay.md)



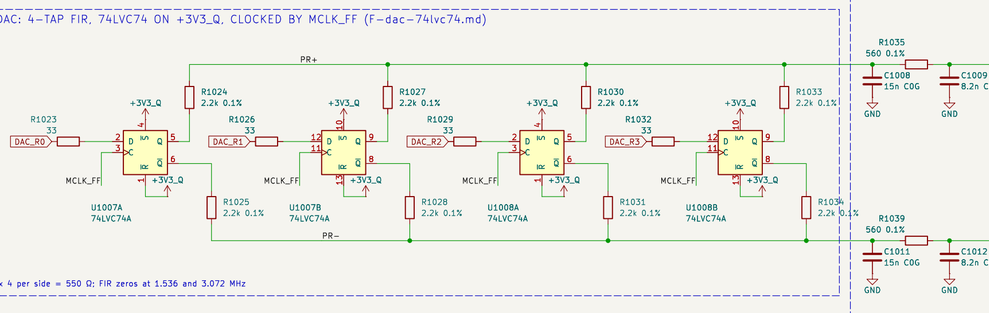
MCLK_Q BUFFER ON +3V3_Q (F-clk-buffer.md)

74LVC74 SUPPLY, 100 nF AT EACH PIN 14

OP-AMP SUPPLIES: 100 nF + 10 µF PER RAIL AT EACH PACKAGE

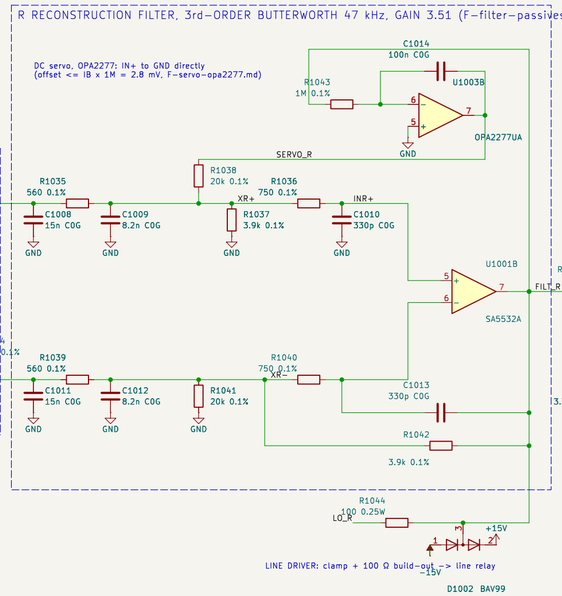
OUTPUT RELAYS, DRIVERS, ESD, JACKS

R 1-BIT DAC: 4-TAP FIR, 74LVC74 ON +3V3_Q, CLOCKED BY MCLK_FF (F-dac-74lvc74.md)

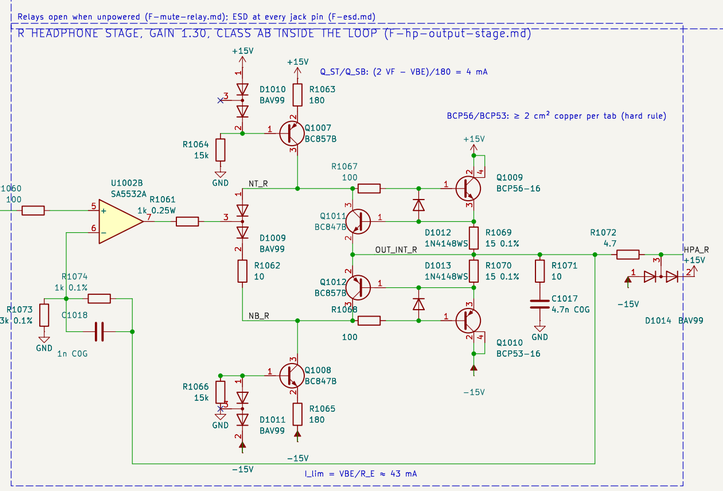


R_T 2.2k x 4 per side = 550 Ω; FIR zeros at 1.536 and 3.072 MHz

R RECONSTRUCTION FILTER, 3rd-ORDER BUTTERWORTH 47 kHz, GAIN 3.51 (F-filter-passives.md)



R HEADPHONE STAGE, GAIN 1.30, CLASS AB INSIDE THE LOOP (F-hp-output-stage.md)

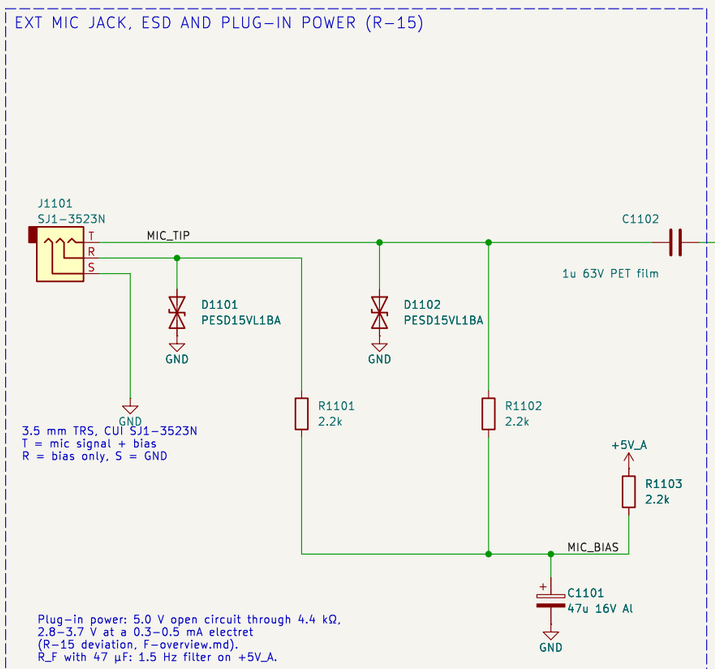


DAC/filter: F-dac-74lvc74.md, F-filter-passives.md; HP: F-hp-output-stage.md
Block F. Audits: design/parts/F-*.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp
Sheet: /audio_out/
File: audio_out.kicad_sch

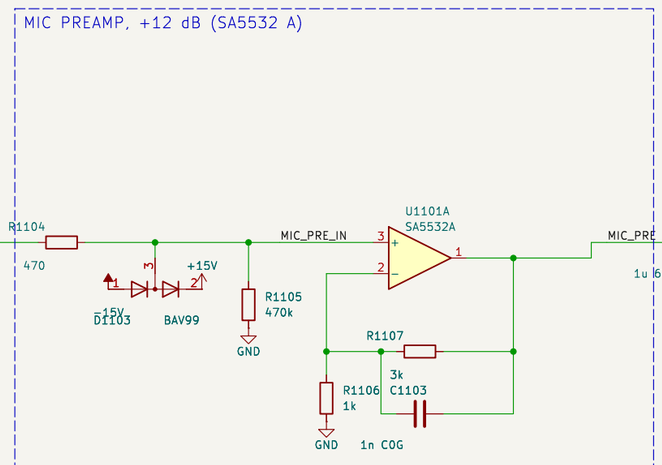
Title: Studio Amp satellite: 1-bit DAC, headphone and line out

Size: A2	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 12/14

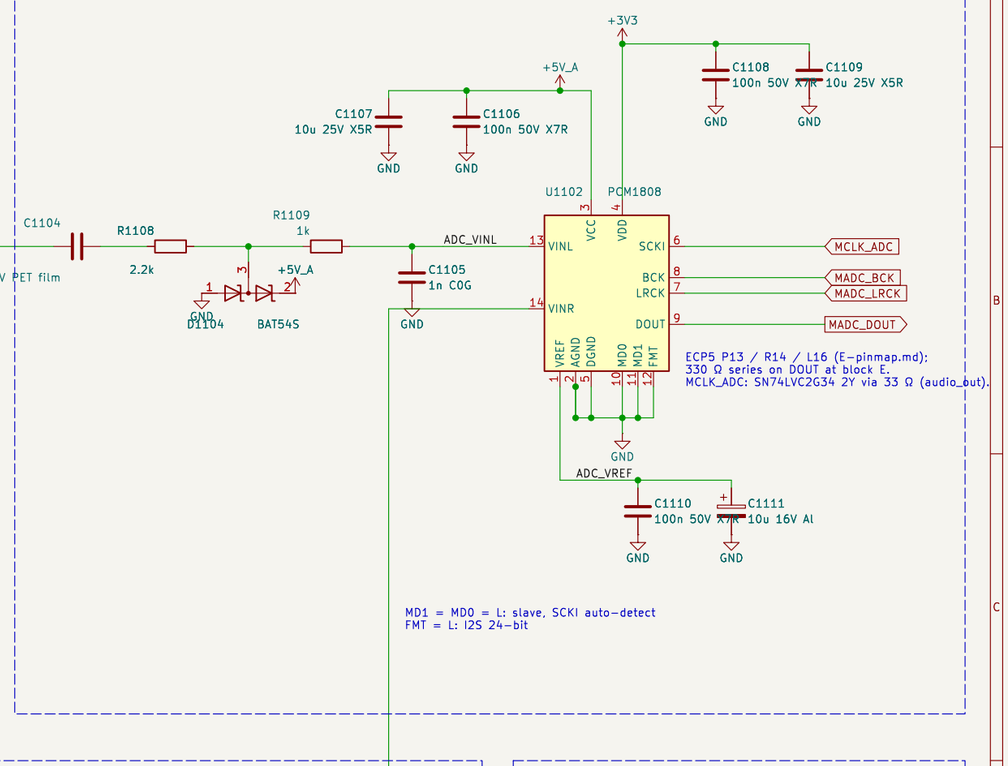
EXT MIC JACK, ESD AND PLUG-IN POWER (R-15)



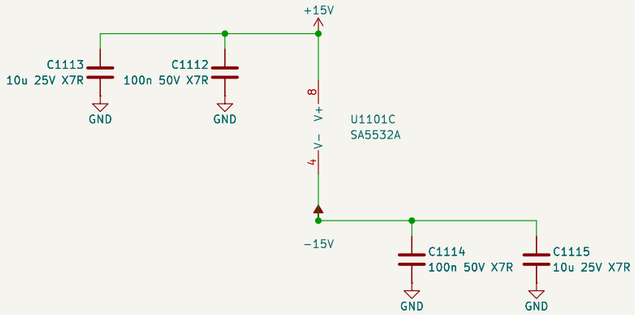
MIC PREAMP, +12 dB (SA5532 A)



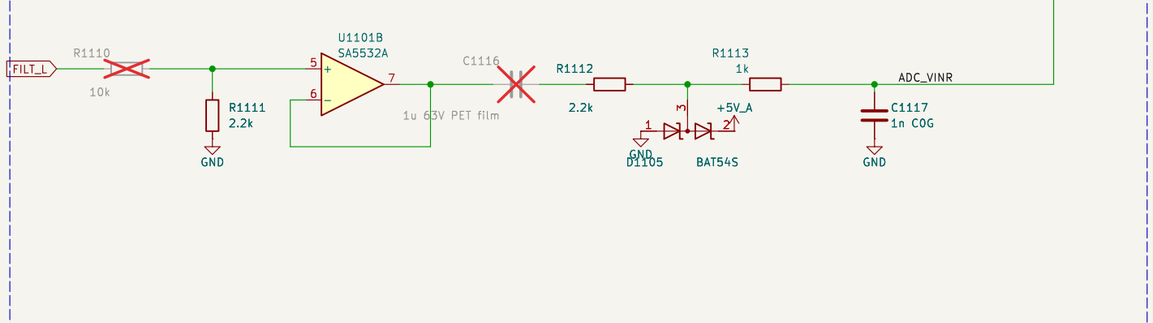
MIC ADC: PCM1808, I2S SLAVE, SCKI = 512 fs



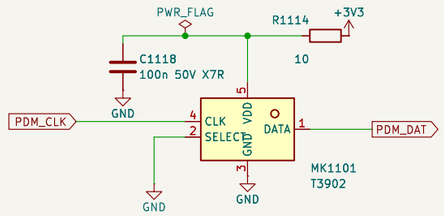
SA5532 SUPPLY (U1101)



LOOPBACK: LINE-OUT L (FILT_L) -> PCM1808 VINR (OPTIONAL, R_LA AND C_LB DNP)



ON-BOARD MEMS MIC (R-16)



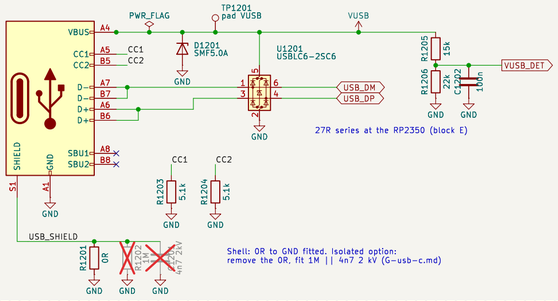
Mic chain: F-mic-adc-pcm1808.md; MEMS mic: F-mems-t3902.md
Block F. Audits: design/parts/F-*.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /audio_in/
File: audio_in.kicad_sch

Title: Studio Amp satellite: Measurement mic and MEMS mic

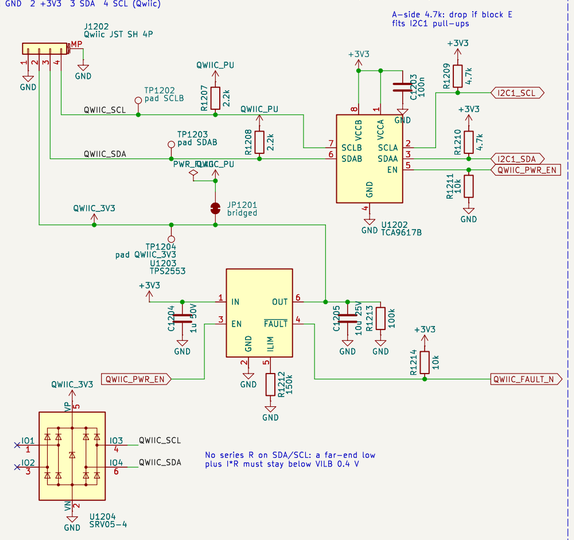
Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 13/14

USB-C USB 2.0 FS to the RP2350, sink only (G-usb-c.md)

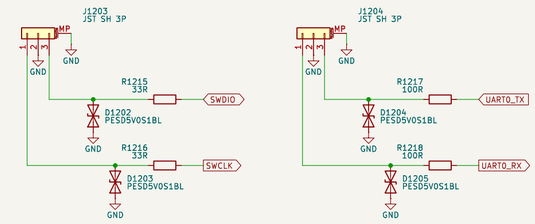


QWIIC TCA9617B repeater + TPS2553 switch, 155–207 mA (G-qwiic.md)

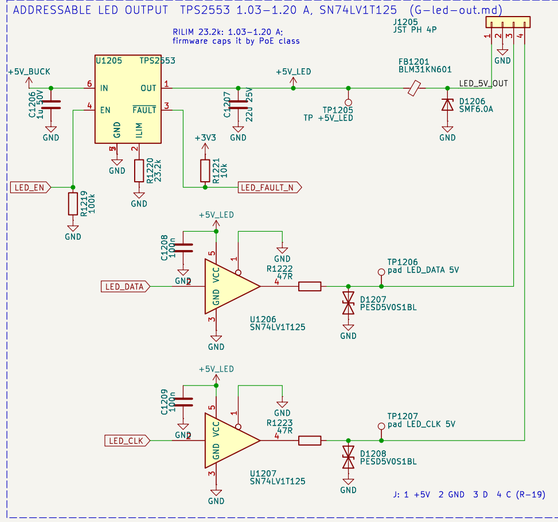
1 GND 2 +3V3 3 SDA 4 SCL (Qwiic)



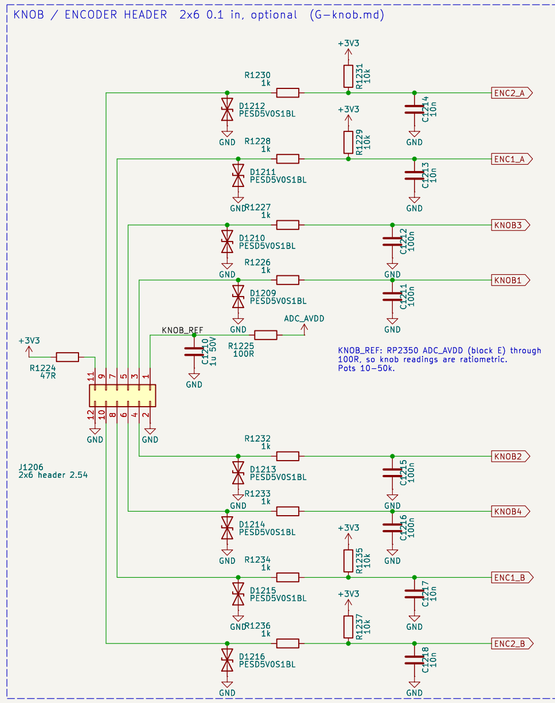
DEBUG PROBE JST SH 3-pin, Sequoia's convention (G-debug-jtag.md)



ADDRESSABLE LED OUTPUT TPS2553 1.03–1.20 A, SN74LV1T125 (G-led-out.md)

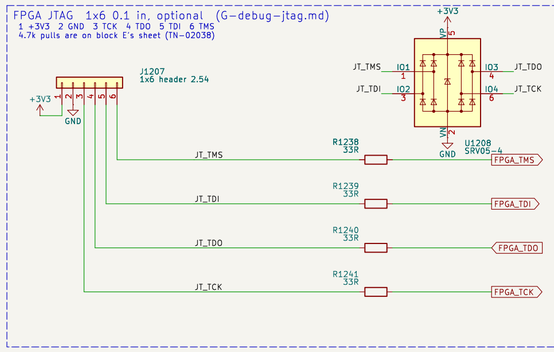


KNOB / ENCODER HEADER 2x6 0.1 in, optional (G-knob.md)



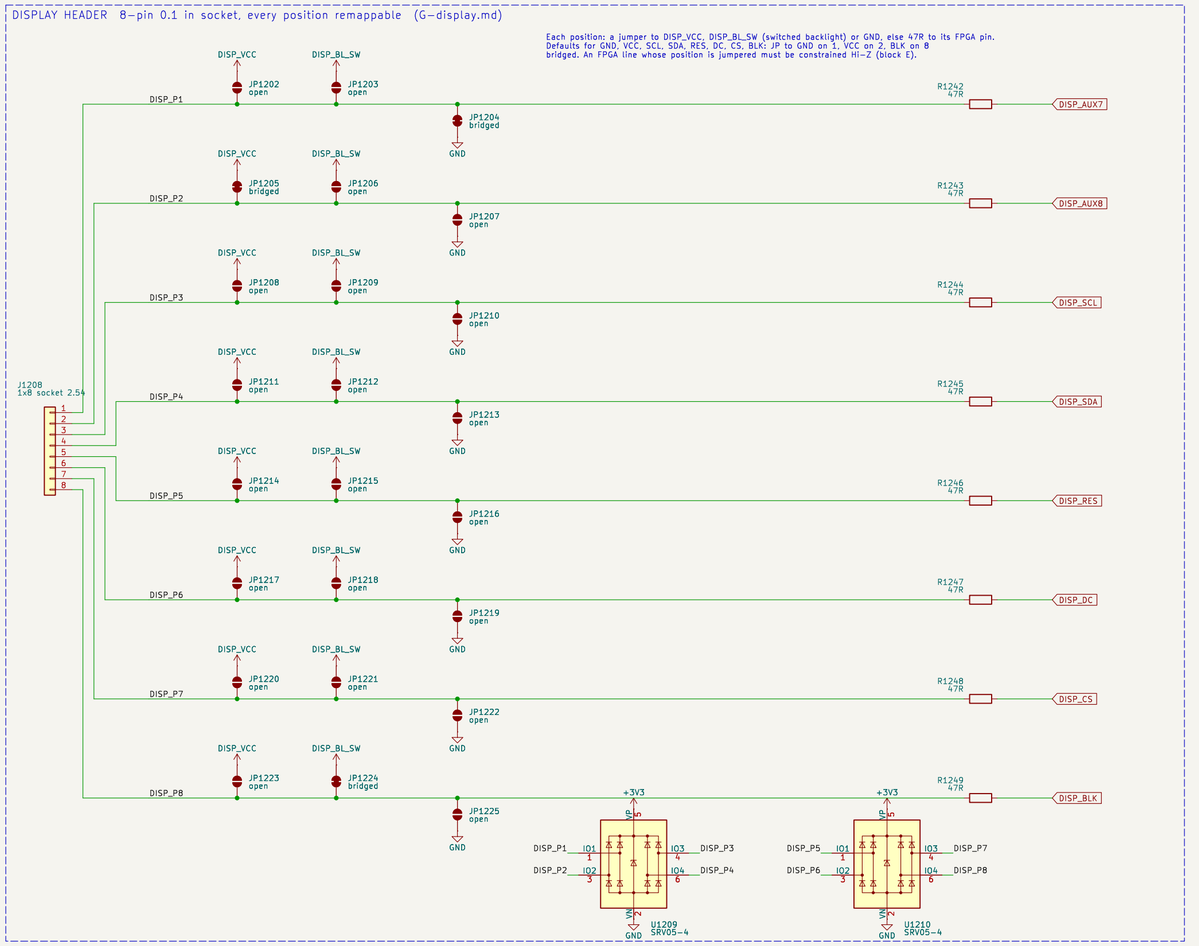
FPGA JTAG 1x6 0.1 in, optional (G-debug-jtag.md)

1 +3V3 2 GND 3 TCK 4 TDO 5 TDI 6 TMS
4.7k pulls are on block E's sheet (TN-02038)

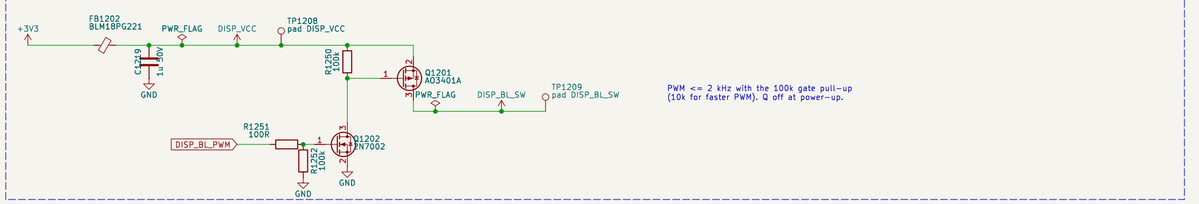


DISPLAY HEADER 8-pin 0.1 in socket, every position remappable (G-display.md)

Each position: a jumper to DISP_VCC, DISP_BL_SW (switched backlight) or GND, else 47R to its FPGA pin.
Defaults for GND, VCC, SCL, SDA, RES, DC, CS, BLK: JP to GND on 1, VCC on 2, BLK on 8
bridged. An FPGA line whose position is jumpered must be constrained HI-Z (block E).



DISP_VCC AND BACKLIGHT SWITCH (G-display.md)



TEST POINTS (R-80, G-testpoints.md) rail pads sit at their rails



Block G. Audits: design/parts/G-*.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp
Sheet: /io/
File: io.kicad_sch

Title: Studio Amp satellite: User I/O, debug and test points

Size: A2	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 14/14